Smart Healthcare Resource Allocation and Appointment Scheduling System

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements

for the Course of

CSA4305 - Internet Programming

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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Module: Hospital Resource Management

Under the Supervision of

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SIMATS ENGINEERING

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SEPTEMBER 2026

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Smart Healthcare Resource Allocation and Appointment Scheduling System" has been carried out by G.Pradeep Reddy (192311289), B.Vishnuvardhan (192311267), M.Reshwanth (192411235), B.Maheshwar (192411195) under the supervision of the project guide and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR	COURSE FACULTY
Dr Marie Claude .N Professor	Dr Sudhagar .D Professor

DECLARATION

We, G.Pradeep Reddy (192311289), B.Vishnuvardhan (192311267), M.Reshwanth (192411235), B.Maheshwar (192411195), of the CSE, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Smart Healthcare Resource Allocation and Appointment Scheduling System" is the result of our own bona fide efforts. The work presented herein is original to the best of our knowledge and has been carried out in accordance with engineering ethics and academic integrity. Sources, references, data, software resources and AI-assisted resources used during the project have been appropriately acknowledged. We further declare that results and findings have not been fabricated or manipulated and that applicable institutional and professional requirements have been followed.

Place: Chennai

Date: 08-09-2026

Signature of the Students with Names

G.Pradeep Reddy (192311289)

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INDIVIDUAL CONTRIBUTION STATEMENT

Student / Register No.	**Specific Responsibilities**	**Design & Development**	**Testing & Analysis**	**Report Contribution**	**Approx. Contribution**
G.Pradeep Reddy (192311289)	Appointment scheduling, authentication and integration	Login, appointment booking, role workflows	Booking and authentication tests	Ch. 1, Ch. 4	25%
B.Vishnuvardhan (192311267)	Resource and bed/room allocation	Resource database and allocation logic	Allocation and conflict tests	Ch. 2, Ch. 3	25%
M.Reshwanth (192411235)	Doctor availability and queue management	Doctor schedules, queues and availability	Scheduling and workload tests	Ch. 3, Ch. 4	25%
B.Maheshwar (192411195)	Dashboard, reports and documentation	Responsive UI, reports and monitoring	UI and report verification	Ch. 4, Ch. 5, Appendices	25%
TOTAL					100%

ABSTRACT

Smart Healthcare Resource Allocation and Appointment Scheduling System is a web-based hospital management application designed to improve the utilization of healthcare resources while reducing appointment delays and scheduling conflicts. Hospitals must coordinate doctors, consultation rooms, beds, diagnostic resources and patient appointments while dealing with changing availability and demand. Manual scheduling can lead to resource underutilization, double booking, long waiting times and inefficient allocation.

The proposed system combines patient appointment scheduling, doctor availability management, hospital resource tracking, allocation rules, conflict detection and administrative monitoring into a single platform. Patients or authorized staff can request appointments based on available time slots, while administrators can manage doctors, rooms and other resources. The system validates availability before confirming an appointment and records allocation information for later review.

The system is organized around major functions: User Authentication and Role Management; Doctor and Resource Management; Appointment Scheduling; Resource Allocation and Conflict Prevention; and Dashboard, Reports and Monitoring. A MySQL database stores users, doctors, resources, schedules, appointments and allocation records, while PHP provides server-side application logic. HTML, CSS, Bootstrap and JavaScript provide the responsive interface.

The engineering design emphasizes efficient resource utilization, prevention of schedule conflicts, role-based access, validation, traceability and usability. The prototype demonstrates how Internet Programming concepts can be applied to a practical healthcare resource management problem in a controlled academic environment.

Keywords: Healthcare Resource Management, Appointment Scheduling, Hospital Resources, Doctor Availability, Conflict Detection, PHP, MySQL

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	**Description**
API	Application Programming Interface
DB	Database
EHR	Electronic Health Record
HTTPS	Hypertext Transfer Protocol Secure
PHP	PHP: Hypertext Preprocessor
SQL	Structured Query Language
UI	User Interface
RBAC	Role-Based Access Control
OPD	Outpatient Department
ICU	Intensive Care Unit
MFA	Multi-Factor Authentication

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Hospitals operate with limited and highly utilized resources such as doctors, consultation rooms, beds, diagnostic facilities and appointment slots. Efficient coordination of these resources is essential for maintaining service quality and reducing patient waiting time. When scheduling is performed manually or through separate systems, appointments may overlap, resources may remain unused, and staff may spend considerable time resolving conflicts.

This project proposes a Smart Healthcare Resource Allocation and Appointment Scheduling System. The system provides a centralized platform for managing hospital resources and scheduling appointments according to availability. It connects users, doctors, schedules and resources through a structured database and validates requests before confirming bookings.

1.2 Need for the Project

- Manual scheduling can result in duplicate appointments and resource conflicts.
- Doctors and rooms may be underutilized when availability is not centrally tracked.
- Patients may experience longer waiting times because appointment slots are not optimally coordinated.
- Hospital administrators need visibility into resource usage and pending appointments.
- A centralized system can improve traceability of appointments, allocations and cancellations.
- A practical academic implementation demonstrates how web technologies can solve a real operational problem.

1.3 Problem Statement

The engineering problem is to design and implement a web-based system that schedules healthcare appointments and allocates hospital resources while balancing availability, waiting time, usability, reliability, security and implementation cost. The system should prevent double booking, verify doctor and resource availability, support cancellations and updates, and maintain records that allow administrators to monitor utilization.

1.4 Project Aim

To develop a secure and user-friendly hospital resource management application that improves appointment scheduling, resource utilization and operational visibility.

1.5 Project Objectives

Design a centralized web platform for managing doctors, hospital resources and appointments.

Maintain doctor schedules and resource availability in a structured database.

Allow authorized users to create, modify and cancel appointments.

Prevent appointment overlap and resource double booking through availability validation.

Allocate suitable rooms, beds or other resources according to predefined rules.

Provide dashboards and reports for appointment status and resource utilization.

Evaluate the prototype using functional and conflict-oriented test cases.

1.6 Scope

The project includes user registration and login, role management, doctor availability, hospital resource records, appointment creation and cancellation, resource allocation, conflict checking, dashboard monitoring and reporting. The prototype is intended for a controlled web-server environment such as XAMPP.

The project does not claim to provide a complete hospital information system, clinical decision support, emergency triage, medical diagnosis, real-time medical device integration or enterprise-scale deployment. These capabilities may be considered as future extensions.

1.7 Expected Engineering Outcome

The expected outcome is a functional web prototype demonstrating the workflow: authentication → availability check → appointment request → resource allocation → confirmation → appointment completion/cancellation → resource release → reporting. The prototype should provide evidence through defined functional and scheduling tests.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review is structured around four areas relevant to the proposed system: healthcare appointment scheduling, resource allocation, availability management and secure web application development. Existing hospital and appointment-management approaches are considered to identify practical requirements for the prototype.

2.2 Review of Existing Work

Appointment systems commonly provide patient registration, provider schedules, appointment slots and reminders. Resource management systems extend this model by tracking rooms, beds, equipment and staff availability. Scheduling approaches can use rule-based allocation to ensure that a requested appointment does not conflict with an existing booking.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual appointment registers	Simple and inexpensive	High risk of overlap and poor visibility	Motivates centralized scheduling
Basic online appointment system	Convenient patient booking	May not manage rooms/resources	Provides booking reference
Separate resource tracking	Tracks resource availability	Scheduling may remain disconnected	Motivates integrated allocation
Large hospital information systems	Comprehensive functionality	Complex and costly for a student prototype	Provides conceptual reference
Proposed system	Appointments and resource allocation in one workflow	Requires careful availability rules	Direct project solution

2.4 Research / Engineering Gap

A simple appointment booking interface does not fully address the operational problem when hospital resources are limited. A booking may be possible in time but impossible because a required doctor or room is unavailable. The proposed work integrates appointment scheduling with resource availability and conflict prevention in a modular web application.

2.5 Proposed Contribution

The project contributes a modular prototype that connects user management, doctor schedules, resource records, appointment booking, allocation validation and monitoring. The design focuses on reducing avoidable scheduling conflicts and improving visibility of resource utilization.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
Patient / User	View available appointment slots and request, modify or cancel appointments according to permissions.
Doctor	View assigned appointments and availability schedule.
Administrator	Manage doctors, rooms/resources, schedules, appointments and utilization records.
Hospital Staff	Allocate and release resources and monitor daily scheduling status.
System Operator	Maintain database, application configuration and secure access.

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
User registration and login	Functional	Valid users can authenticate successfully.
Doctor availability	Functional	Available time slots are displayed correctly.
Appointment booking	Functional	Valid appointment requests can be confirmed.
Conflict prevention	Functional	Overlapping bookings are rejected.
Resource allocation	Functional	Required resources are assigned only when available.
Cancellation / release	Functional	Cancelled appointments release allocated resources.
Role-based access	Non-Functional	Unauthorized functions are denied.
Responsive UI	Non-Functional	Core pages remain usable on desktop and mobile-sized screens.
Input validation	Non-Functional	Invalid form data is rejected.

3.2 Constraints & Applicable Standards / Practices

Standard / Practice	Requirement	Application
OWASP Web Security Practices	Address common web risks	Validation, sessions, authorization and secure coding

HTTPS / TLS	Protect data in transit	Recommended for deployment
SQL prepared statements	Reduce injection risk	Parameterized database queries
Password hashing	Protect stored credentials	Passwords stored as hashes
RBAC	Restrict functions by role	Separate patient/staff/admin capabilities

3.3 Design Specifications

Parameter	**Target / Requirement**	**Priority**
Database	MySQL	High
Backend	PHP	High
Appointment validation	Doctor + time + resource availability	High
Resource tracking	Room/bed/equipment status	High
Conflict handling	Reject overlapping allocation	High
Session management	Server-side authenticated sessions	High
Interface	Responsive Bootstrap UI	Medium
Reports	Appointment and resource utilization summaries	Medium

3.4 Alternative Solutions & Evaluation

Criterion	**Manual / Paper**	**Basic Online Booking**	**Integrated Proposed System**
Cost	High staff effort	Medium	Medium
Scheduling visibility	Low	Medium	High
Resource allocation	Low	Low/Medium	High
Conflict prevention	Low	Medium	High
Traceability	Low	Medium	High

3.5 Selected Design & Detailed Design

A PHP and MySQL architecture is selected because it is suitable for the course environment and can be deployed using XAMPP. The browser communicates with the PHP application layer. Authentication and role checks are performed before sensitive operations. Appointment requests are validated against doctor schedules and resource availability before confirmation. The database stores users, doctors, resources, schedules, appointments and allocation records.

System Architecture:

Browser / User Interface → PHP Application Layer → Authentication & Authorization → Scheduling & Allocation Logic → MySQL Database

3.6 Trade-off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
Rule-based allocation	Manual assignment	Consistent conflict checking	Needs clear rules	Selected
PHP + MySQL	Node.js + MongoDB	Fits course/XAMPP environment	Less flexible for some architectures	Selected
Server-side sessions	JWT-only authentication	Simple browser integration	Needs careful session configuration	Selected
Centralized database	Separate spreadsheets	Consistent data and queries	Requires database maintenance	Selected

3.7 Sustainability, Ethics, Safety & Security

Domain	Consideration	Impact on Final Decision
Sustainability	Avoid unnecessary duplicated records and manual processing.	Centralized digital workflow.
Ethics	Protect patient and staff information.	Role-based access and minimum necessary data exposure.
Safety	Avoid treating scheduling software as a clinical decision system.	No diagnosis or clinical decision support in scope.
Security	Protect accounts and operational records.	Hashing, sessions, validation, prepared statements and access checks.

Risk Register

Risk	Likelihood	Severity	Mitigation	Residual Risk
Double booking	Medium	High	Availability validation before confirmation	Low
Incorrect resource status	Medium	High	Controlled update and release workflow	Medium
Unauthorized access	Medium	High	Authentication and role checks	Low
SQL injection	Medium	High	Prepared statements	Low
Incorrect user input	Medium	Medium	Validation and constrained fields	Low
Server failure	Low	Medium	Backups and recovery procedu	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows an incremental development approach. User authentication and database connectivity are implemented first, followed by doctor/resource management. Appointment scheduling and conflict validation are then integrated, followed by dashboards, reporting and testing. Module-level and integration testing are performed before evaluating the complete workflow.

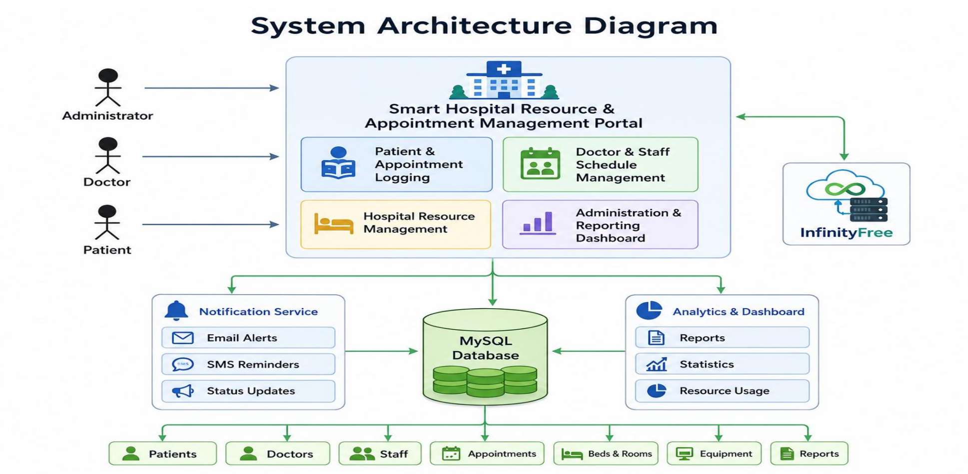

Fig 3.1

Resource	Purpose
XAMPP	Local Apache and MySQL development environment.
PHP	Server-side application logic.
MySQL	User, doctor, resource, appointment and allocation storage.
HTML/CSS/Bootstrap	Responsive web interface.
JavaScript/AJAX	Client-side interaction and asynchronous requests where needed.
Git	Source-code version control and collaboration.

4.2 Software Implementation & Modern Tools Used

Tool / Software	Purpose	Stage Used
PHP	Backend application logic	All implementation stages
MySQL	Persistent data storage	Database design and integration
Bootstrap 5	Responsive UI	Frontend development
JavaScript	Client-side validation and interaction	Frontend development
XAMPP	Apache/PHP/MySQL local server	Development and testing

Git	Version control	Development and integration

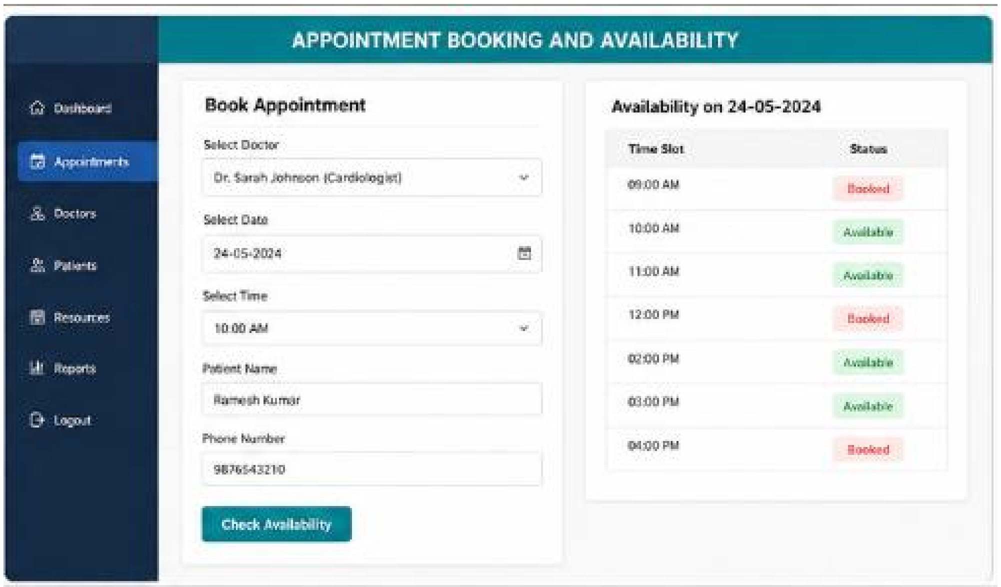

Fig. 3.2 Appointment Booking and Availability

Fig.3.2

4.3 Module-wise Implementation

Module 1 – User Authentication & Role Management

- Registration validates required information and stores a password hash.
- Login verifies credentials and creates an authenticated session.
- Role checks restrict administrative and scheduling functions.
- Logout terminates the active session.

Module 2 – Doctor & Hospital Resource Management

- Administrators can register doctors and maintain their availability schedules.
- Rooms, beds and other hospital resources can be recorded with status information.
- Resource status is updated when resources are allocated, released or marked unavailable.

Module 3 – Appointment Scheduling

- Users select a suitable doctor and available time slot.
- The system checks for overlapping appointments before confirmation.
- Appointment status can be tracked as scheduled, completed or cancelled.
- Cancellation updates the corresponding resource availability.

Module 4 – Resource Allocation, Dashboard & Reports

- The system allocates an appropriate available resource for a confirmed appointment.
- Allocation conflicts are rejected rather than silently overwriting existing bookings.
- Dashboards summarize appointments, resource status and utilization.
- Administrators can review scheduling records for operational monitoring.

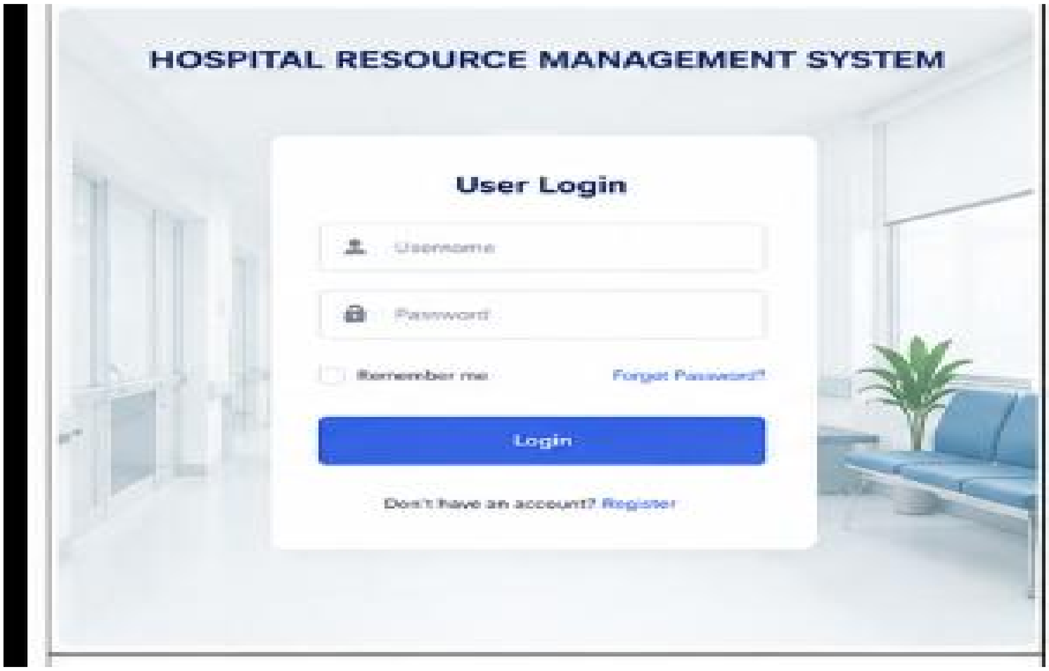

Fig.4.1

4.4 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Registration	Submit valid and invalid forms	Valid user created; invalid data rejected
Login	Use correct and incorrect credentials	Correct credentials authenticate; incorrect credentials denied
Doctor availability	Create schedule and query slots	Only valid available slots are shown
Appointment	Book a free slot	Appointment is created successfully
Conflict prevention	Book an already occupied slot	Second conflicting booking is rejected
Resource allocation	Allocate occupied and free resources	Only available resources are assigned
Cancellation	Cancel a confirmed appointment	Appointment cancelled and resource released
Role security	Access restricted functions using wrong role	Unauthorized action denied
Dashboard	Perform sample transactions	Counts and status values update correctly

4.5 Results

The intended prototype results are evaluated against functional requirements and scheduling controls. The final implementation should demonstrate that appointment conflicts are detected, resources are allocated only when available, cancellations release resources, and role restrictions are enforced. Screenshots of login, dashboard, appointment booking, resource allocation and administrative monitoring should be included as implementation evidence.

Result Area	Expected Observation	Actual Result
Authentication	Valid users log in and invalid users are rejected.	To be recorded
Availability	Only available doctors/resources appear for booking.	To be recorded
Appointment	Valid appointments are created without overlap.	To be recorded
Conflict control	Overlapping appointment/resource requests are rejected.	To be recorded
Resource release	Cancelled/completed bookings release resources.	To be recorded
Dashboard	Operational status is reflected in summaries.	To be recorded

Note: Actual achieved values and Pass/Fail status should be filled after executing the final implementation test cycle. The tables above define verification criteria rather than claiming unperformed measurements.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Develop healthcare scheduling system	Integrated PHP/MySQL web prototype	To be validated
Maintain doctor/resource availability	Doctor and resource modules	To be validated
Prevent scheduling conflicts	Availability and overlap checks	To be validated
Allocate hospital resources	Allocation workflow and resource status	To be validated
Provide monitoring	Dashboard and reports	To be validated

4.7 Strengths & Limitations

Strengths:

- Appointments and resource availability are managed in one workflow.
- Conflict detection reduces duplicate scheduling.
- Resource status can be monitored and updated.
- Role-based access separates administrative functions.
- The architecture is modular and suitable for an academic Internet Programming project.

Limitations:

- The prototype depends on correct resource and schedule data entered by authorized staff.
- Enterprise-scale hospital integration is outside the current scope.
- Real-time integration with medical devices and clinical systems is not implemented.
- Advanced optimization algorithms are not part of the initial prototype.
- Final performance values depend on the actual deployment environment and workload.

Fig.4.2

4.8 Project Planning, Roles & Budget

Phase	Activity	Planned Duration
1	Requirement analysis and literature review	Week 1
2	Database and system design	Week 2
3	Authentication and role management	Week 3
4	Doctor/resource management	Week 4
5	Appointment scheduling and conflict checks	Week 5
6	Allocation, dashboards and reports	Week 6
7	Testing and verification	Week 7
8	Documentation and final presentation	Week 8

Student	Assigned Responsibility	Actual Contribution
G.Pradeep Reddy	Appointment scheduling, authentication and integration	To be confirmed
B.Vishnuvardhan	Resource and bed/room allocation	To be confirmed
M.Reshwanth	Doctor availability and queue management	To be confirmed

B.Maheshwar	UI, reports and documentation	To be confirmed

Item	Estimated Cost	Actual Cost
Development software	₹0	₹0 / To be confirmed
XAMPP / PHP / MySQL	₹0	₹0
Domain / hosting (optional)	₹0–2,000	To be confirmed
Miscellaneous	₹500	To be confirmed
Total	₹500–2,500	To be confirmed

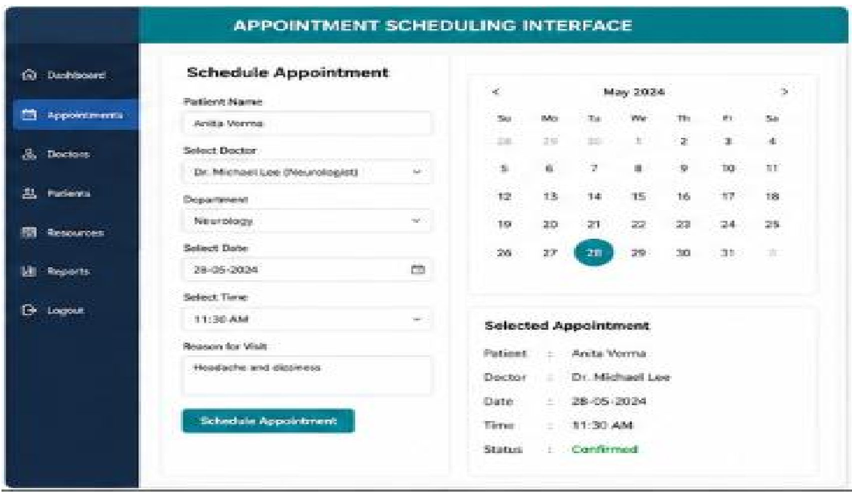

Fig. 4.3 Appointment Scheduling Interface

Fig.4.3

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Smart Healthcare Resource Allocation and Appointment Scheduling System is designed to address operational challenges associated with hospital appointment scheduling and limited healthcare resources. The proposed architecture combines authentication, doctor availability, resource tracking, appointment booking, conflict prevention, allocation and monitoring within a single web application.

T resources are allocated, and cancellations or completions release resources for subsequent use. The approach demonstrates how web development and database concepts can be applied to improve the coordination of healthcare operations.

5.2 Future Work

- Introduce optimization algorithms to improve resource utilization and reduce waiting time.
- Add SMS, email or mobile notifications for appointment reminders.
- Integrate calendar synchronization for doctors and departments.
- Add real-time dashboards for hospital administrators.
- Support multiple departments and more complex resource dependencies.
- Integrate with electronic health record systems through secure APIs where appropriate.
- Introduce multi-factor authentication and stronger audit controls.
- Develop a mobile application for patients and staff.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical application of appointment scheduling and resource allocation logic.
- Database relationships for users, doctors, resources, schedules and appointments.
- Role-based authentication and secure web application practices.
- Testing of scheduling conflicts and operational workflows.
- Integration of frontend, backend and database components.

Engineering & Professional Skills Developed:

- Requirements analysis and modular system design.
- Backend integration and database design.
- Software testing, debugging and documentation.
- Team coordination, version control and technical communication.

Individual Reflection – G.Pradeep Reddy: The main challenge was coordinating authentication with appointment operations. The work improved understanding of role-based access and the importance of validating scheduling requests before confirmation.

Individual Reflection – B.Vishnuvardhan: The main challenge was representing hospital resources and their availability. The work improved understanding of allocation rules, status tracking and conflict prevention.

Individual Reflection – M.Reshwanth: The main challenge was maintaining doctor schedules and appointment timing without overlaps. The work improved understanding of scheduling logic, availability queries and cancellation handling.

Individual Reflection – B.Maheshwar: The main challenge was integrating the dashboard, reports and documentation into a coherent workflow. The work improved skills in responsive UI development, testing and technical reporting.

REFERENCES

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2. OWASP Foundation, OWASP Top 10: The Ten Most Critical Web Application Security Risks.
3. OWASP Foundation, OWASP Cheat Sheet Series – Authentication, Session Management and Input Validation.
4. PHP Documentation, PHP Manual – Password Hashing, Sessions, Database Access and Web Application Development.
5. MySQL Documentation, MySQL Reference Manual – SQL Statements, Prepared Statements and Database Security.
6. Mozilla Developer Network (MDN), Web security, HTTP, HTML, CSS and JavaScript documentation.
7. General literature on healthcare appointment scheduling, resource allocation and hospital operations management.
8. Note: The final submitted report should acknowledge the exact software versions, repositories, libraries, datasets (if any), and AI-assisted resources actually used during implementation.

APPENDICES

Appendix A – Appointment Scheduling Workflow

Authenticate the user and establish a secure session.

Load doctor schedules and hospital resource availability.

Select a valid appointment date and time.

Check doctor availability and existing appointments.

Check required resource availability.

Create the appointment and allocation records.

Update resource status where required.

Allow cancellation or completion and release the resource.

Record relevant operational activity.

Appendix B – Database Design

Table	Important Fields	Purpose
users	id, name, email, password_hash, role, created_at	User identity and authentication
doctors	id, name, department, specialization, status	Doctor information
resources	id, resource_name, type, location, status	Hospital resource records
doctor_schedule	id, doctor_id, date, start_time, end_time	Doctor availability
appointments	id, patient_id, doctor_id, date, start_time, end_time, status	Appointment records
allocations	id, appointment_id, resource_id, allocated_at, released_at	Resource allocation
audit_logs	id, user_id, action, timestamp, details	Operational auditing

Appendix C – Source Code / Code Repository Information

Repository URL: ____

Appendix D – Screenshots / Implementation Evidence

- Login and registration page.
- Hospital resource dashboard.
- Doctor availability page.
- Appointment scheduling page.
- Resource allocation page.
- Appointment status / cancellation result.
- Administrative monitoring dashboard.

Appendix E – Experimental / Raw Data

Record actual test results for appointment booking time, conflict detection, resource allocation, cancellation, dashboard updates and failure cases after the final prototype is executed. Do not insert estimated measurements as achieved results.

Appendix F – Ethics / Institutional Approval

The prototype should use synthetic or non-confidential data for demonstrations unless appropriate institutional approval and data-handling procedures are available. Real patient identifiers and confidential clinical information should not be included in screenshots or demonstration datasets.

Appendix G – Project Meeting / Progress Records

Date	Discussion / Task	Decision / Action	Member(s)
________	Requirement discussion	Finalize module structure	________
________	Design review	Approve architecture	________
________	Implementation review	Integrate modules	________
________	Testing review	Resolve defects	________

Appendix H – User / Industry Feedback

To be completed if feedback is collected from users, faculty, industry mentors or other evaluators.

Appendix I – Publications / Patents / Competitions / Product Outcomes

To be completed if the project produces a publication, patent, competition entry or product outcome.

Appendix J – Individual Contribution Evidence

Attach repository commits, module ownership records, screenshots of implementation work, testing records and report sections contributed by each team member.